```
"C:\Users\nicop\OneDrive\Documentos\Universidad\Proyecto de Grado STREDEMIA\STREDEMIA\.venv\Scripts\python.exe" "C:\Users\nicop\OneDrive\Documentos\Universidad\Proyecto de Grado STREDEMIA\STREDEMIA\.venv\Modelos\DNN_Model.py"
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776285904.270305   23420 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776285907.569020   23420 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
Entrenamiento: (381, 33)
Validación:    (37, 33)
Prueba:        (37, 33)
Variables de entrada (features): 33
  - Ítems SISCO:     21
  - Variables contextuales: 12
Scaler guardado en carpeta 'Modelos'

inicio de validación cruzada con 5 folds (pliegues)...

*-*-*-* Fold 1/5
I0000 00:00:1776285909.546808   23420 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
F1-score en fold 1: 0.9417
*-*-*-* Fold 2/5
F1-score en fold 2: 0.8925
*-*-*-* Fold 3/5
WARNING:tensorflow:5 out of the last 7 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x0000022DCAC92FC0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
WARNING:tensorflow:6 out of the last 9 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x0000022DCAC92FC0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
F1-score en fold 3: 0.9410
*-*-*-* Fold 4/5
F1-score en fold 4: 0.8569
*-*-*-* Fold 5/5
F1-score en fold 5: 0.9881

*-*-*-* Resumen validación cruzada *-*-*-*-*
  Fold 1: F1-macro = 0.9417
  Fold 2: F1-macro = 0.8925
  Fold 3: F1-macro = 0.9410
  Fold 4: F1-macro = 0.8569
  Fold 5: F1-macro = 0.9881
  Promedio: 0.9240
  Desv. estándar: 0.0452

*-*-*-*- Entrenamiento de modelo final *-*-*-*-*-
Epoch 1/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 22ms/step - accuracy: 0.4952 - loss: 0.9883 - val_accuracy: 0.7568 - val_loss: 0.8749
Epoch 2/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.6794 - loss: 0.7345 - val_accuracy: 0.7838 - val_loss: 0.7667
Epoch 3/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.7656 - loss: 0.5770 - val_accuracy: 0.7568 - val_loss: 0.6855
Epoch 4/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.7943 - loss: 0.4822 - val_accuracy: 0.7838 - val_loss: 0.5850
Epoch 5/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8493 - loss: 0.4004 - val_accuracy: 0.7568 - val_loss: 0.5100
Epoch 6/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8852 - loss: 0.3139 - val_accuracy: 0.7297 - val_loss: 0.4731
Epoch 7/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8876 - loss: 0.3213 - val_accuracy: 0.7568 - val_loss: 0.4378
Epoch 8/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9043 - loss: 0.2888 - val_accuracy: 0.8649 - val_loss: 0.4041
Epoch 9/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9163 - loss: 0.2253 - val_accuracy: 0.8919 - val_loss: 0.3612
Epoch 10/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9306 - loss: 0.2261 - val_accuracy: 0.8649 - val_loss: 0.3310
Epoch 11/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9258 - loss: 0.1991 - val_accuracy: 0.9189 - val_loss: 0.2713
Epoch 12/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9402 - loss: 0.1840 - val_accuracy: 0.9189 - val_loss: 0.2846
Epoch 13/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9330 - loss: 0.1651 - val_accuracy: 0.9189 - val_loss: 0.2808
Epoch 14/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9474 - loss: 0.1706 - val_accuracy: 0.9459 - val_loss: 0.1931
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Epoch 15/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9426 - loss: 0.1434 - val_accuracy: 0.9189 - val_loss: 0.2313
Epoch 16/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9689 - loss: 0.1344 - val_accuracy: 0.9189 - val_loss: 0.2128
Epoch 17/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9474 - loss: 0.1390 - val_accuracy: 0.9189 - val_loss: 0.1775
Epoch 18/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9665 - loss: 0.1042 - val_accuracy: 0.9459 - val_loss: 0.1785
Epoch 19/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9569 - loss: 0.1066 - val_accuracy: 0.9459 - val_loss: 0.1410
Epoch 20/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9641 - loss: 0.1262 - val_accuracy: 0.9459 - val_loss: 0.1648
Epoch 21/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9593 - loss: 0.1217 - val_accuracy: 0.9459 - val_loss: 0.1816
Epoch 22/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9593 - loss: 0.1033 - val_accuracy: 0.9459 - val_loss: 0.1489
Epoch 23/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9785 - loss: 0.0801 - val_accuracy: 0.9730 - val_loss: 0.1275
Epoch 24/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9833 - loss: 0.0519 - val_accuracy: 0.9730 - val_loss: 0.1169
Epoch 25/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9761 - loss: 0.0696 - val_accuracy: 0.9459 - val_loss: 0.1436
Epoch 26/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9641 - loss: 0.0890 - val_accuracy: 0.9730 - val_loss: 0.0743
Epoch 27/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9833 - loss: 0.0674 - val_accuracy: 0.9730 - val_loss: 0.0662
Epoch 28/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9785 - loss: 0.0740 - val_accuracy: 0.9730 - val_loss: 0.0526
Epoch 29/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9833 - loss: 0.0509 - val_accuracy: 0.9730 - val_loss: 0.0903
Epoch 30/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9737 - loss: 0.1043 - val_accuracy: 0.9730 - val_loss: 0.0792
Epoch 31/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9809 - loss: 0.0426 - val_accuracy: 0.9730 - val_loss: 0.0622
Epoch 32/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9761 - loss: 0.0637 - val_accuracy: 0.9730 - val_loss: 0.0423
Epoch 33/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9928 - loss: 0.0378 - val_accuracy: 0.9730 - val_loss: 0.0345
Epoch 34/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9809 - loss: 0.0493 - val_accuracy: 1.0000 - val_loss: 0.0255
Epoch 35/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9761 - loss: 0.0984 - val_accuracy: 0.9730 - val_loss: 0.0381
Epoch 36/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9761 - loss: 0.0532 - val_accuracy: 0.9730 - val_loss: 0.0338
Epoch 37/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9880 - loss: 0.0434 - val_accuracy: 1.0000 - val_loss: 0.0219
Epoch 38/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9976 - loss: 0.0321 - val_accuracy: 1.0000 - val_loss: 0.0123
Epoch 39/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9904 - loss: 0.0312 - val_accuracy: 1.0000 - val_loss: 0.0164
Epoch 40/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9952 - loss: 0.0220 - val_accuracy: 0.9730 - val_loss: 0.0278
Epoch 41/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9856 - loss: 0.0454 - val_accuracy: 1.0000 - val_loss: 0.0143
Epoch 42/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9856 - loss: 0.0481 - val_accuracy: 1.0000 - val_loss: 0.0033
Epoch 43/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9928 - loss: 0.0262 - val_accuracy: 1.0000 - val_loss: 0.0140
Epoch 44/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9952 - loss: 0.0290 - val_accuracy: 0.9730 - val_loss: 0.0229
Epoch 45/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9928 - loss: 0.0183 - val_accuracy: 1.0000 - val_loss: 0.0029
Epoch 46/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9880 - loss: 0.0322 - val_accuracy: 1.0000 - val_loss: 0.0032
Epoch 47/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9856 - loss: 0.0436 - val_accuracy: 1.0000 - val_loss: 0.0049
Epoch 48/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9880 - loss: 0.0357 - val_accuracy: 1.0000 - val_loss: 0.0031
Epoch 49/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9928 - loss: 0.0204 - val_accuracy: 1.0000 - val_loss: 0.0098
Epoch 50/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9904 - loss: 0.0346 - val_accuracy: 1.0000 - val_loss: 0.0104
Epoch 51/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9928 - loss: 0.0152 - val_accuracy: 1.0000 - val_loss: 0.0078
Epoch 52/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9928 - loss: 0.0166 - val_accuracy: 1.0000 - val_loss: 0.0015
Epoch 53/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9976 - loss: 0.0150 - val_accuracy: 1.0000 - val_loss: 0.0014
Epoch 54/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9952 - loss: 0.0105 - val_accuracy: 1.0000 - val_loss: 0.0059
Epoch 55/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9880 - loss: 0.0409 - val_accuracy: 1.0000 - val_loss: 0.0030
Epoch 56/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9904 - loss: 0.0258 - val_accuracy: 1.0000 - val_loss: 0.0075
Epoch 57/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9856 - loss: 0.0412 - val_accuracy: 1.0000 - val_loss: 0.0050
Epoch 58/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9952 - loss: 0.0241 - val_accuracy: 1.0000 - val_loss: 0.0020
Epoch 59/150
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9904 - loss: 0.0301 - val_accuracy: 1.0000 - val_loss: 0.0061
Epoch 60/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9928 - loss: 0.0228 - val_accuracy: 1.0000 - val_loss: 0.0032
Epoch 61/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9952 - loss: 0.0147 - val_accuracy: 1.0000 - val_loss: 0.0015
Epoch 62/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9976 - loss: 0.0129 - val_accuracy: 1.0000 - val_loss: 0.0017
Epoch 63/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9976 - loss: 0.0103 - val_accuracy: 1.0000 - val_loss: 0.0015
Epoch 64/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9952 - loss: 0.0210 - val_accuracy: 1.0000 - val_loss: 8.0156e-04
Epoch 65/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9928 - loss: 0.0211 - val_accuracy: 1.0000 - val_loss: 0.0014
Epoch 66/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9928 - loss: 0.0285 - val_accuracy: 1.0000 - val_loss: 0.0016
Epoch 67/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9833 - loss: 0.0291 - val_accuracy: 1.0000 - val_loss: 8.3318e-04
Epoch 68/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9976 - loss: 0.0126 - val_accuracy: 1.0000 - val_loss: 0.0011
Epoch 69/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9952 - loss: 0.0174 - val_accuracy: 1.0000 - val_loss: 0.0011
Epoch 70/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9976 - loss: 0.0095 - val_accuracy: 1.0000 - val_loss: 9.2364e-04
Epoch 71/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9952 - loss: 0.0157 - val_accuracy: 1.0000 - val_loss: 8.9947e-04
Epoch 72/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 1.0000 - loss: 0.0089 - val_accuracy: 1.0000 - val_loss: 7.8971e-04
Epoch 73/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9928 - loss: 0.0116 - val_accuracy: 1.0000 - val_loss: 6.8964e-04
Epoch 74/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9928 - loss: 0.0168 - val_accuracy: 1.0000 - val_loss: 0.0014
Epoch 75/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9952 - loss: 0.0143 - val_accuracy: 1.0000 - val_loss: 8.4426e-04
Epoch 76/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9976 - loss: 0.0071 - val_accuracy: 1.0000 - val_loss: 9.4550e-04
Epoch 77/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9952 - loss: 0.0080 - val_accuracy: 1.0000 - val_loss: 8.0049e-04
Epoch 78/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 1.0000 - loss: 0.0083 - val_accuracy: 1.0000 - val_loss: 3.0901e-04
Epoch 79/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9976 - loss: 0.0074 - val_accuracy: 1.0000 - val_loss: 2.5346e-04
Epoch 80/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9976 - loss: 0.0072 - val_accuracy: 1.0000 - val_loss: 2.4938e-04
Epoch 81/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9952 - loss: 0.0110 - val_accuracy: 1.0000 - val_loss: 3.3356e-04
Epoch 82/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9928 - loss: 0.0208 - val_accuracy: 1.0000 - val_loss: 0.0013
Epoch 83/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9904 - loss: 0.0253 - val_accuracy: 1.0000 - val_loss: 0.0011
Epoch 84/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9976 - loss: 0.0111 - val_accuracy: 1.0000 - val_loss: 5.0630e-04
Epoch 85/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9928 - loss: 0.0162 - val_accuracy: 1.0000 - val_loss: 9.6257e-04
Epoch 86/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 1.0000 - loss: 0.0065 - val_accuracy: 1.0000 - val_loss: 7.8870e-04
Epoch 87/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9976 - loss: 0.0106 - val_accuracy: 1.0000 - val_loss: 4.7038e-04
Epoch 88/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9976 - loss: 0.0076 - val_accuracy: 1.0000 - val_loss: 4.5462e-04
Epoch 89/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 1.0000 - loss: 0.0038 - val_accuracy: 1.0000 - val_loss: 5.7028e-04
Epoch 90/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 1.0000 - loss: 0.0039 - val_accuracy: 1.0000 - val_loss: 5.1276e-04
Epoch 91/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9976 - loss: 0.0074 - val_accuracy: 1.0000 - val_loss: 4.2732e-04
Epoch 92/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9928 - loss: 0.0171 - val_accuracy: 1.0000 - val_loss: 5.9318e-04
Epoch 93/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9952 - loss: 0.0179 - val_accuracy: 1.0000 - val_loss: 7.4242e-04
Epoch 94/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9976 - loss: 0.0081 - val_accuracy: 1.0000 - val_loss: 7.9112e-04
Epoch 95/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9976 - loss: 0.0059 - val_accuracy: 1.0000 - val_loss: 7.3572e-04

*-*-*-*- Evaluación final sobre datos de prueba *-*-*-*-

Reporte de clasificación:
              precision    recall  f1-score   support

        bajo       1.00      1.00      1.00         3
       medio       0.90      0.93      0.91        28
        alto       0.60      0.50      0.55         6

    accuracy                           0.86        37
   macro avg       0.83      0.81      0.82        37
weighted avg       0.86      0.86      0.86        37

Modelo entrenado con éxito...

```

```
*-*-*-* MÉTRICAS Y EVALUACIONES *-*-*-*-*

AUC-ROC
Macro (one-vs-rest): 0.9347

Por clases:
  AUC-ROC bajo: 1.0000
  AUC-ROC medio: 0.9008
  AUC-ROC alto: 0.9032
Curvas ROC por clases archivadas en carpeta 'Resultados'
Matriz de confusión almacenada en carpeta 'Resultados'
Curvas de entrenamiento guardadas en carpeta 'Resultados'

Modelo ejecutado con éxito.
Métricas y evaluaciones generadas con éxito
Modelo Guardado (Carpeta Modelos)

Process finished with exit code 0

```